

Lecture 4

Stein's unbiased risk estimator

Model Selection

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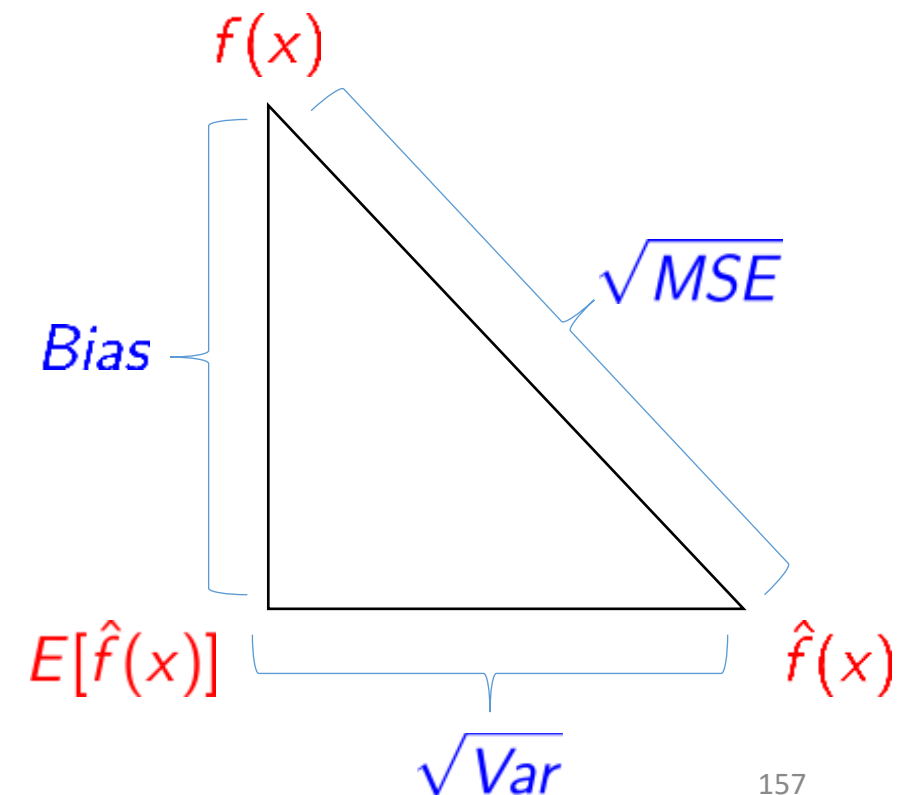
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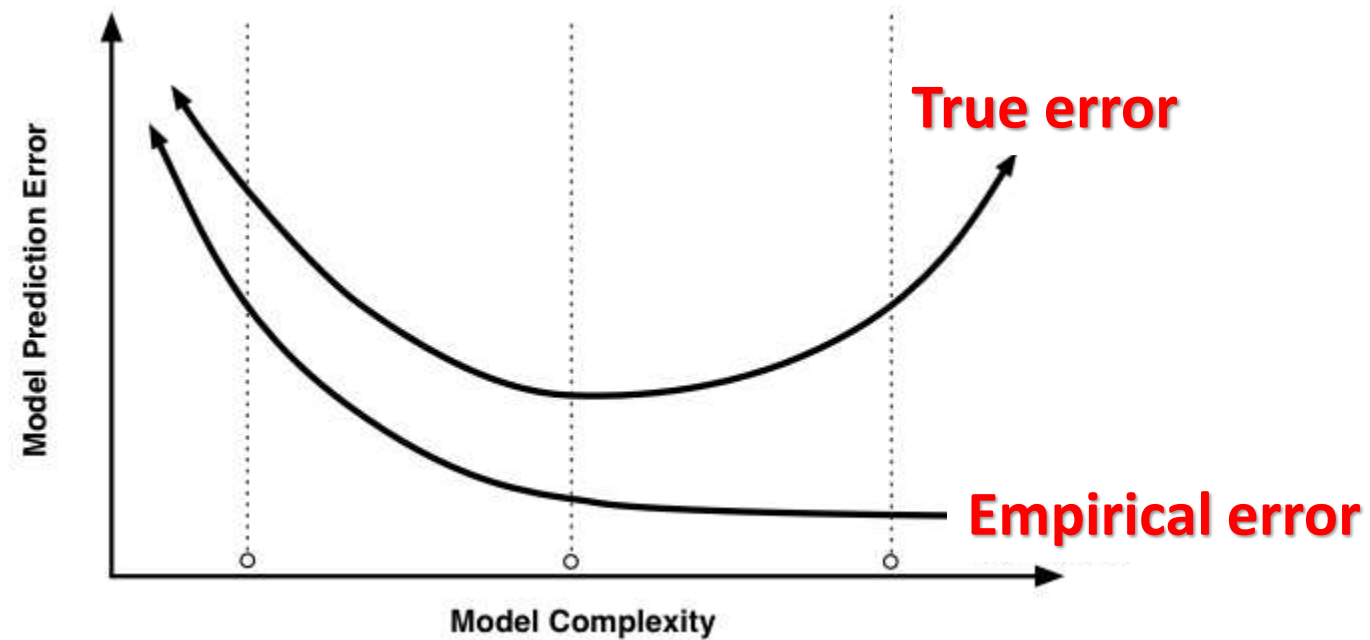
$$\hat{y}_i = \hat{f}(x_i)$$

$$f_i \equiv f(x_i)$$

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- Empirical error:

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



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$$\Rightarrow E[\varepsilon_i (\hat{f}_i - f_i)] = E[(y_i - f_i)(\hat{f}_i - f_i)] = 0$$

If summing up all m points that are not in T .

$$\underbrace{\sum_{i=1}^m (\hat{y}_i - y_i)^2}_{\text{err}} = \underbrace{\sum_{i=1}^m (\hat{f}_i - f_i)^2}_{\text{Err}} + m\sigma^2$$

true error

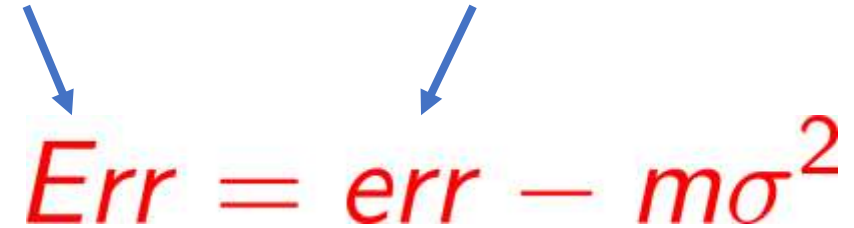
empirical error



$$Err = err - m\sigma^2$$

true error

empirical error


$$Err = err - m\sigma^2$$

Empirical error (*err*) is a good estimator of true error (*Err*) if the point (x_0, y_0) is not in the training set.

case 2

assume: $(x_0, y_0) \in T$

then: $2E[\epsilon_0(\hat{f}_0 - f_0)] \neq 0$

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Stein's Lemma

If :

$$x \sim \mathcal{N}(\theta, \sigma^2)$$

and $g(x)$ is differentiable.

then

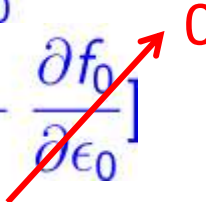
$$E[g(x)(x - \theta)] = \sigma^2 E\left[\frac{\partial g(x)}{\partial x}\right]$$

- Our problem: $E[\epsilon_0(\hat{f}_0 - f_0)] = \sigma^2 E\left[\frac{\partial(\hat{f}_0 - f_0)}{\partial \epsilon_0}\right]$

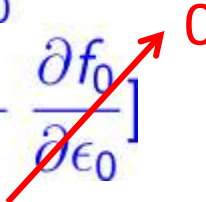
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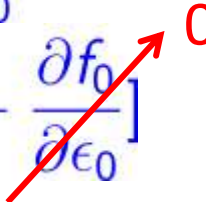
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0

Chain rule

- Our problem:

$$\begin{aligned}
 E[\epsilon_0(\hat{f}_0 - f_0)] &= \sigma^2 E\left[\frac{\partial(\hat{f}_0 - f_0)}{\partial \epsilon_0}\right] \\
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 \end{aligned}$$

Diagram annotations:

- A red arrow labeled **0** points from the term $\frac{\partial f_0}{\partial \epsilon_0}$ in the second line to the first line.
- A blue arrow points from a box labeled **Chain rule** to the term $\frac{\partial \hat{f}_0}{\partial \epsilon_0}$ in the third line.
- A red arrow labeled **1** points from the term $\frac{\partial y_0}{\partial \epsilon_0}$ in the fourth line to the third line.

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 &= \sigma^2 E\left[\frac{\partial \hat{f}_0}{\partial y_0}\right]
 \end{aligned}$$

Diagram illustrating the simplification of the expectation of the derivative of the error term $\epsilon_0(\hat{f}_0 - f_0)$ with respect to ϵ_0 .

The first term $\frac{\partial f_0}{\partial \epsilon_0}$ is crossed out with a red arrow labeled 0, indicating it is zero.

The second term $\frac{\partial \hat{f}_0}{\partial \epsilon_0}$ is simplified using the chain rule, shown as $\frac{\partial \hat{f}_0}{\partial y_0} \cdot \frac{\partial y_0}{\partial \epsilon_0}$. The term $\frac{\partial y_0}{\partial \epsilon_0}$ is crossed out with a red arrow labeled 1, indicating it is 1.

The final result is $\sigma^2 E\left[\frac{\partial \hat{f}_0}{\partial y_0}\right]$.

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- A red bracket under the term $\frac{\partial \hat{f}_0}{\partial y_0}$ in the fifth line is labeled **D_0** .

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Sum over all n data points:

$$\sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (\hat{f}_i - f_i)^2 + n\sigma^2 - 2\sigma^2 \sum_{i=1}^n D_i$$

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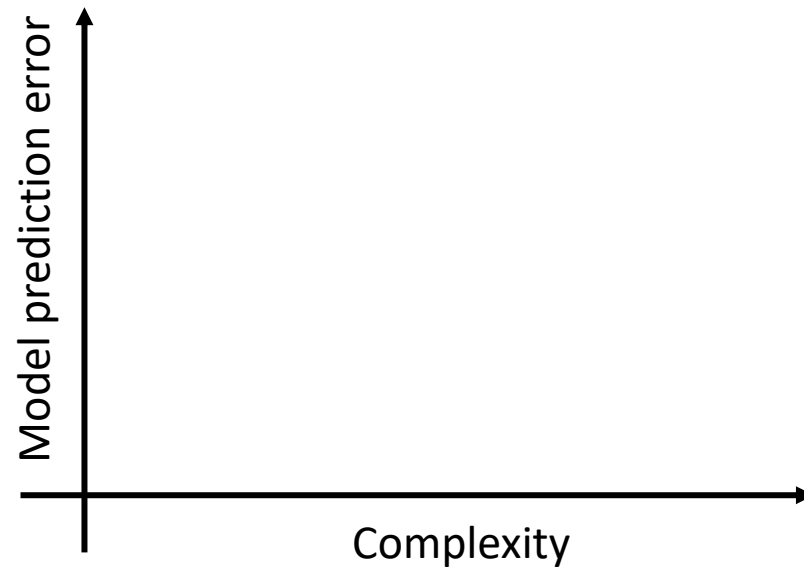
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Complexity of model

Stein's Unbiased Risk Estimator (SURE)

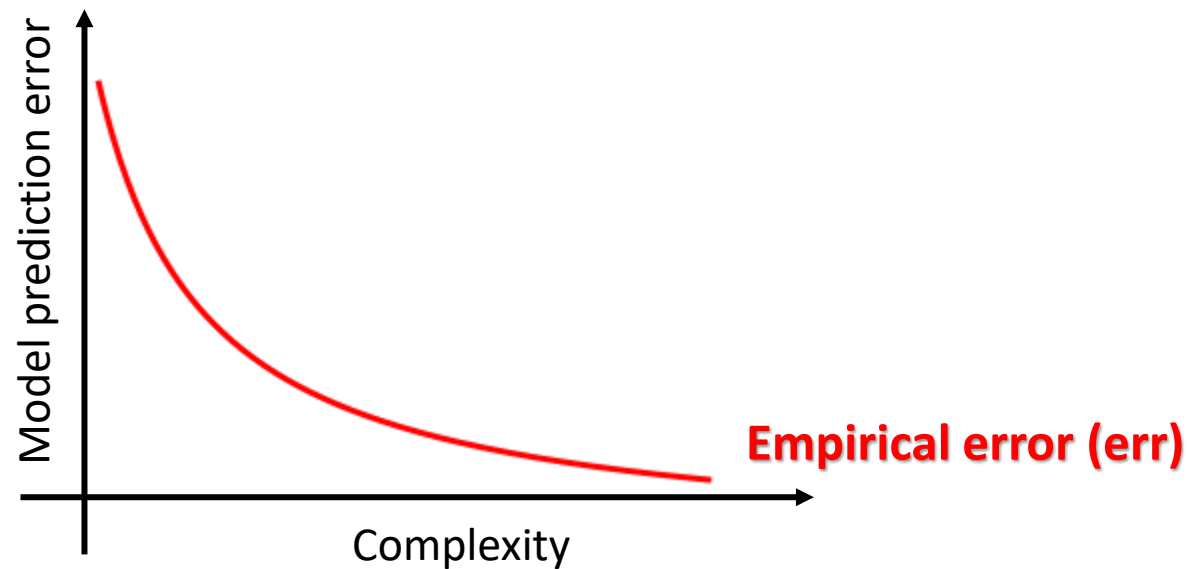
- SURE gives us a very good insight about the behavior of true error

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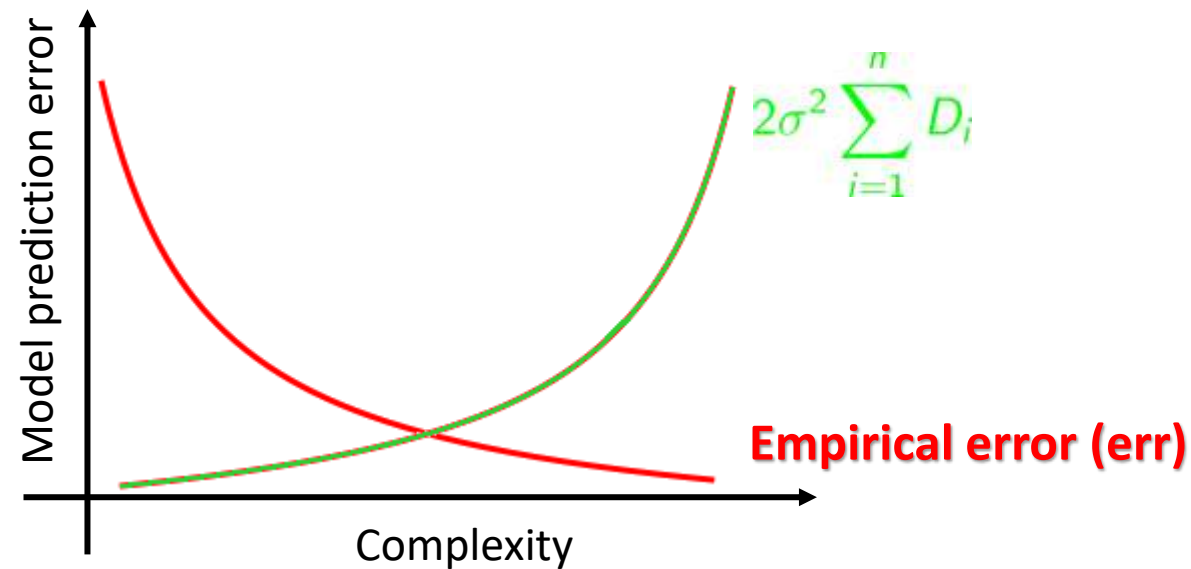
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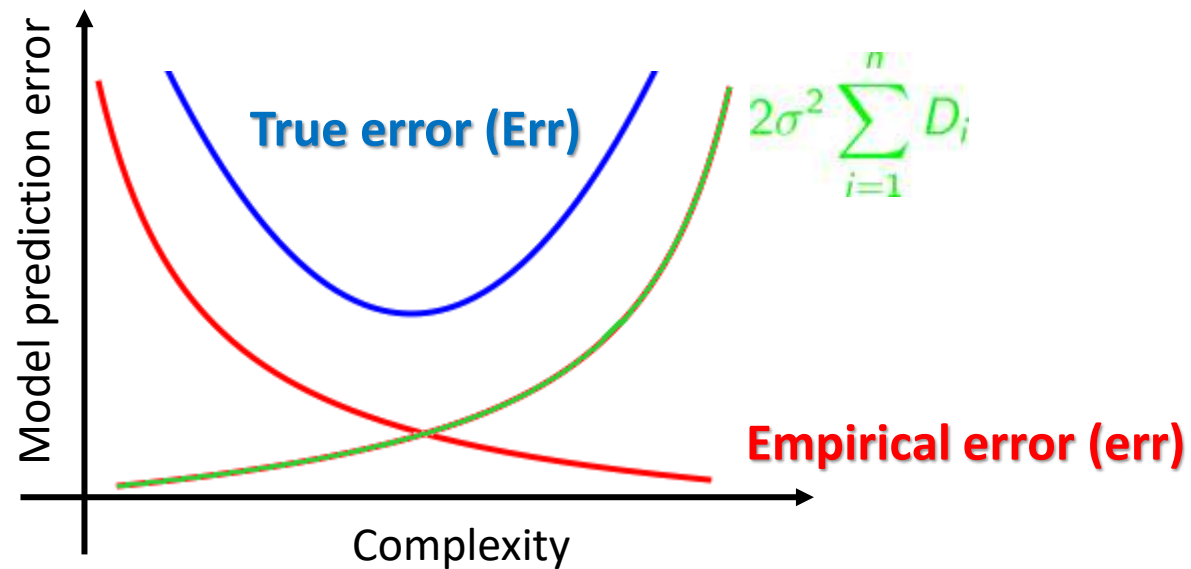
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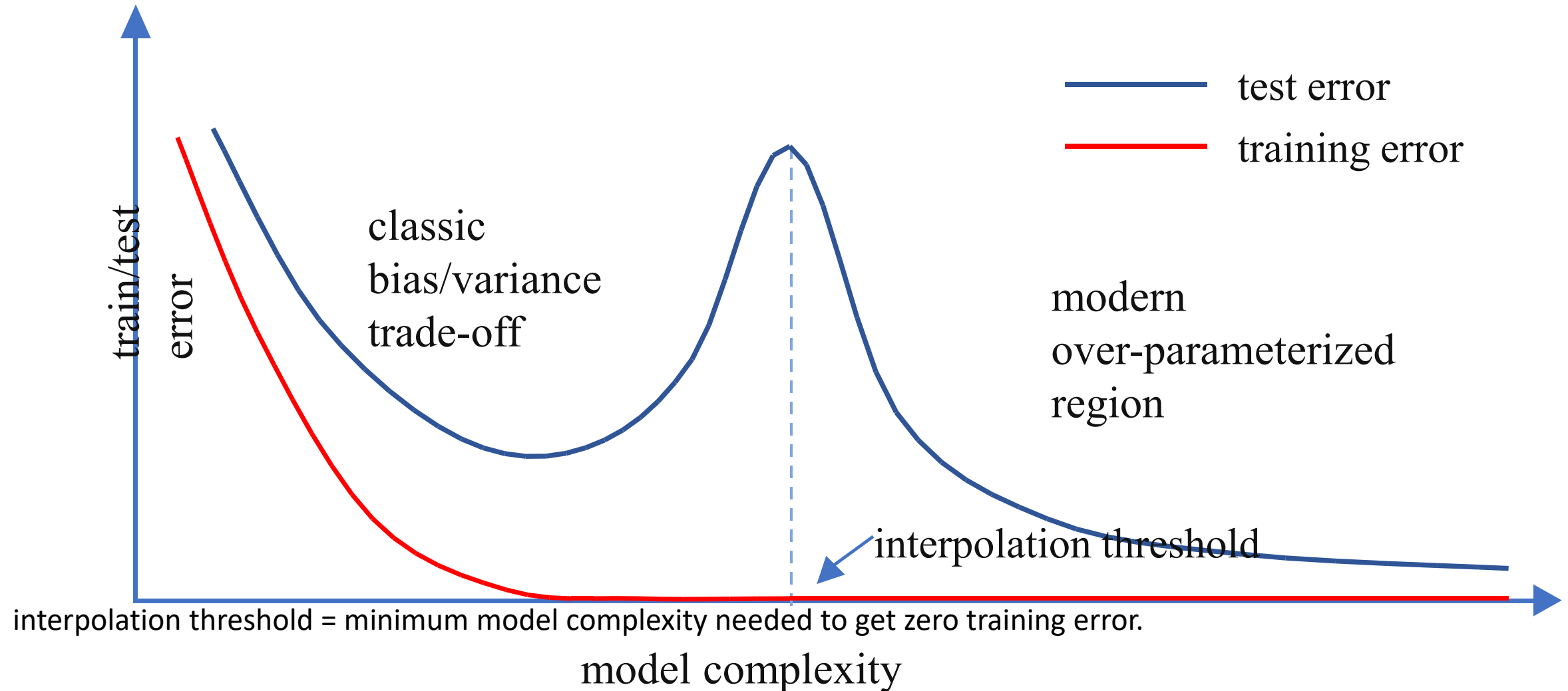


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Double descent: error can decrease again after interpolation



Double descent

- interpolation threshold: minimum model complexity needed to get zero training error.
- As model complexity increases beyond interpolation threshold, there are many more zero-training-error solutions available, and the learning algorithm can often select solutions with better implicit regularization, causing test error to fall again.
- For example, in a regression problem with n training points and p parameters, the interpolation threshold is often around $p \approx n$, though the exact location depends on the model, data, loss, and optimization method.

Implicit Regularization in Overparameterized Linear Regression

Training data:

$$X \in \mathbb{R}^{d \times n}, \quad \mathbf{y} \in \mathbb{R}^n$$

Linear model:

$$\mathbf{y} = \mathbf{w}^\top X$$

with

$$\mathbf{w} \in \mathbb{R}^d$$

Overparameterized regime:

$$d > n$$

Many parameter vectors interpolate the training data:

$$\mathbf{w}^\top X = \mathbf{y}$$

$\implies$ many zero-training-error solutions

Which Solution Does Gradient Descent Choose?

Start from

$$\mathbf{w}_0 = \mathbf{0}$$

Gradient descent converges to the minimum-norm interpolating solution:

$$\mathbf{w}^\star = \arg \min_{\mathbf{w}} \|\mathbf{w}\|_2 \quad \text{subject to} \quad \mathbf{w}^\top X = \mathbf{y}^\top$$

Equivalently,

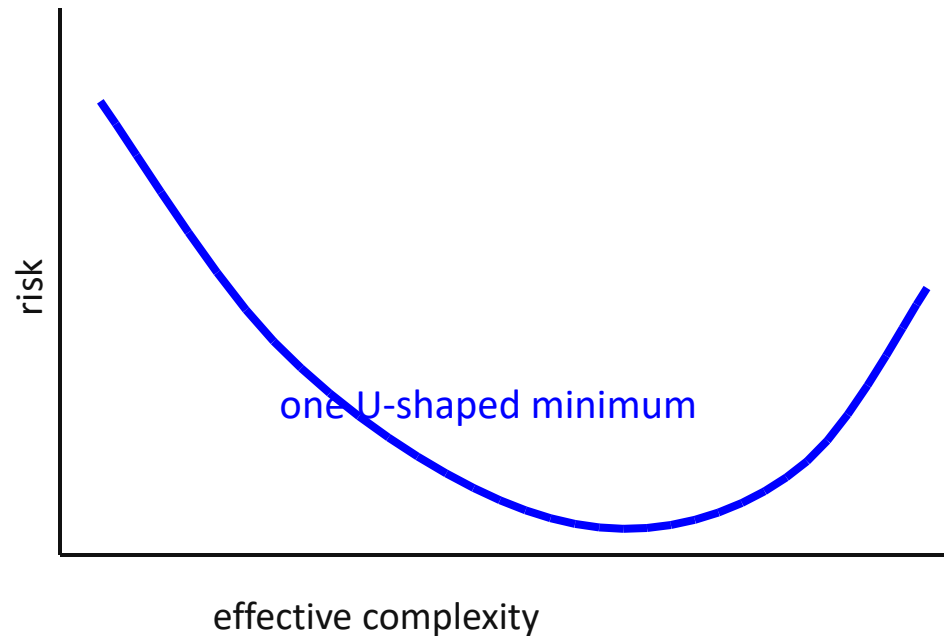
$$\mathbf{w}^\star = X(X^\top X)^{-1}\mathbf{y}$$

No explicit regularizer is added.

The optimization dynamics themselves select a particular interpolating solution: implicit regularization.

SURE estimates risk; it does not explain double descent

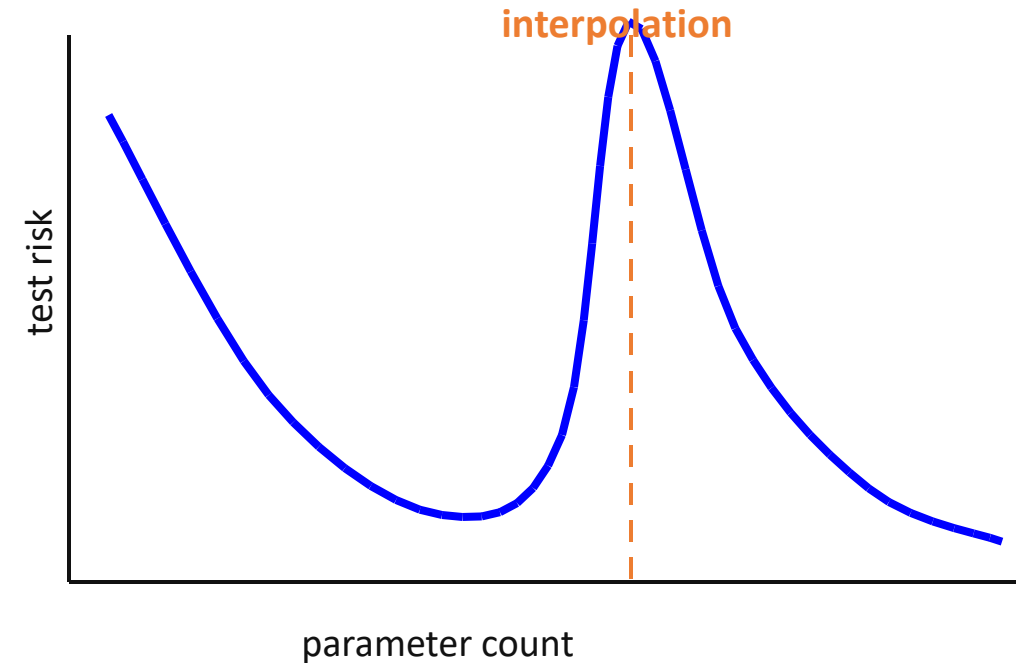
SURE / classical view



$$\text{Err} = \text{err} - n\sigma^2 + 2\sigma^2 \sum_{i=1}^n D_i$$
$$\text{err} = \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

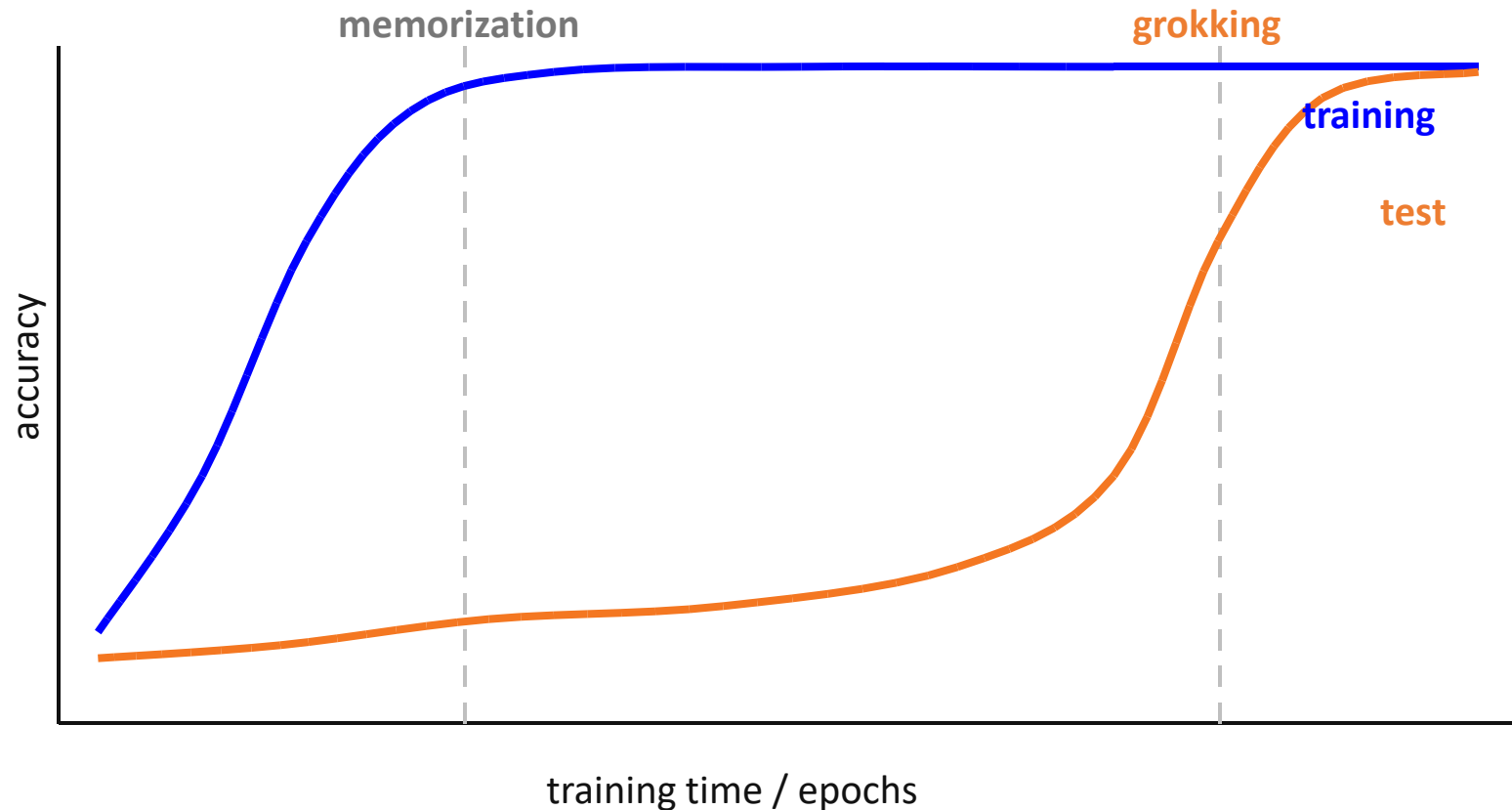
SURE can evaluate a fitted estimator; it does not explain the mechanism producing the second descent.

Double-descent view



Parameter count may not track effective complexity. Optimization also selects among many exact-fit solutions.

Grokking: memorization first, generalization much later



1 Learn examples

Training accuracy rises quickly.

2 Memorize

Training accuracy $\approx 100\%$, while test accuracy remains low.

3 Delayed generalization

Much later, test accuracy rises although model size is unchanged.

Not every overfit model groks.
Power et al. (2022)

SURE does not explain grokking's delayed transition

time t_1 : memorized

time t_2 : generalized



same model size

training error ≈ 0

test error high

continued optimization

same model size

training error ≈ 0

test error low

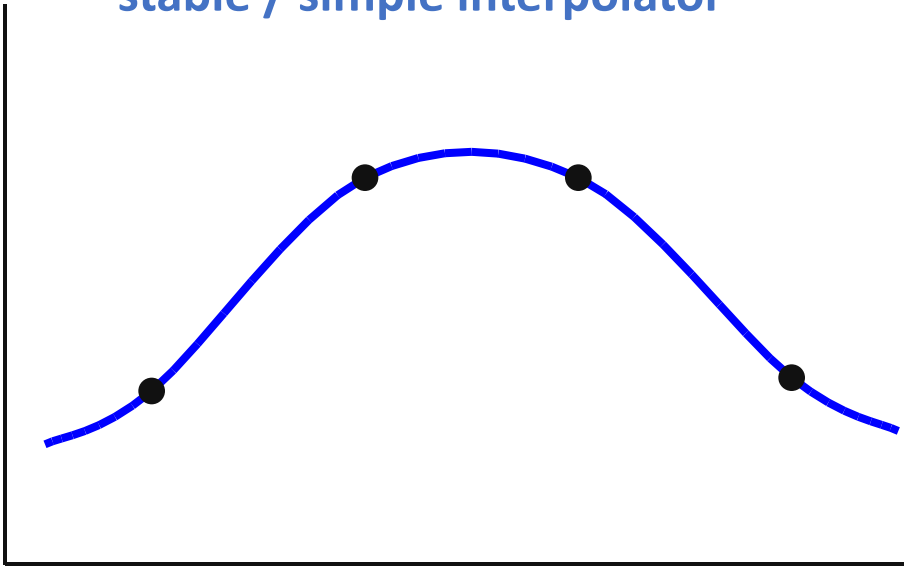
representation • norm •
margin

These solution properties change although parameter count and training error do not.

SURE can estimate risk at t_1 and t_2 ; it does not explain the path from t_1 to t_2 .

Zero training error does not automatically mean harmful overfitting

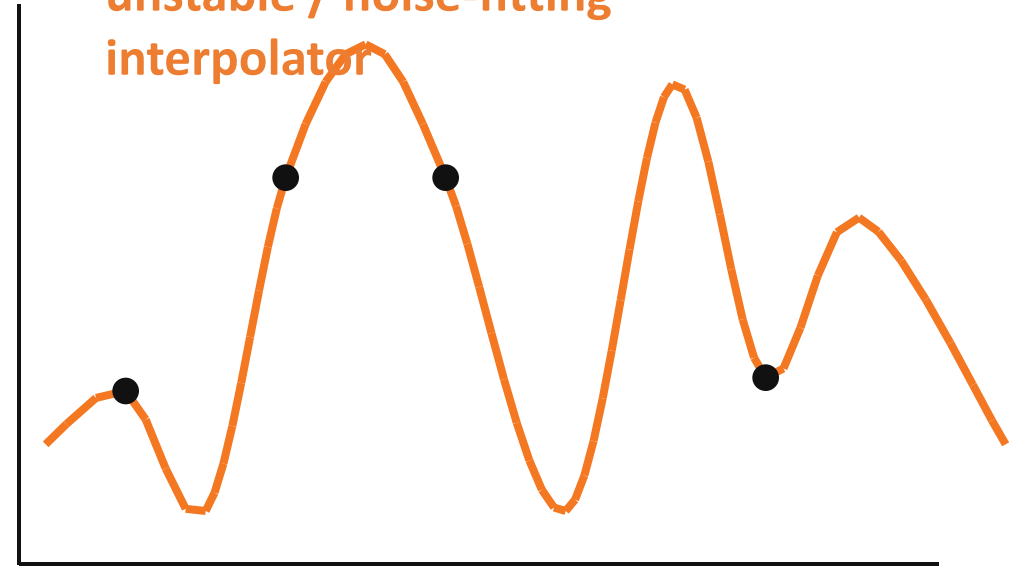
stable / simple interpolator



training error = 0

smooth between examples $\rightarrow$ lower test error

unstable / noise-fitting interpolator



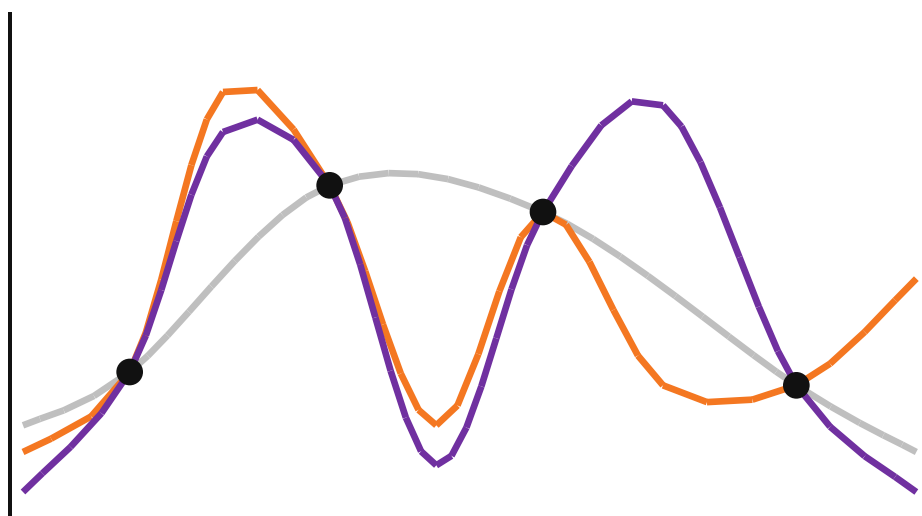
training error = 0

large changes between examples $\rightarrow$ higher test error

Interpolation is a training-data property. Harmful overfitting means poor performance on unseen data.

Why regularization still matters after double descent

many solutions fit the training data

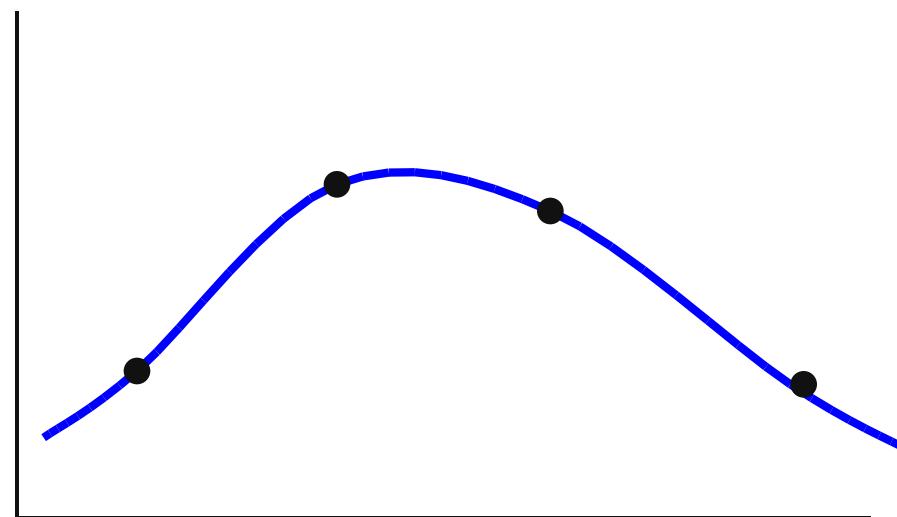


all have training error = 0

regularization

L2 • early stopping
augmentation • noise

preferred solution



lower norm • smoother •
more stable

Regularization does not merely prevent fitting; it chooses which fitting solution we trust.

Goal: low expected test risk